Decoding Gender Assignment in an Opaque System: Experimental Evidence from Czech Nonwords

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1 Introduction

- Czech exhibits a **richly inflected** and relatively **opaque grammatical gender** (GG) system.
- 4 categories: masculine animate/inanimate, feminine, neuter.
- **Phono-morphological cues often inconsistent**: koled**a** "carol"[F], předsed**a** "chairman"[M],srs**t** "fur"[F], prs**t** "finger"[M].
- General ending-to-GG regularities exist, and children use gender features to predict upcoming words (Smolik & Bláhová, 2019).

2 Aim

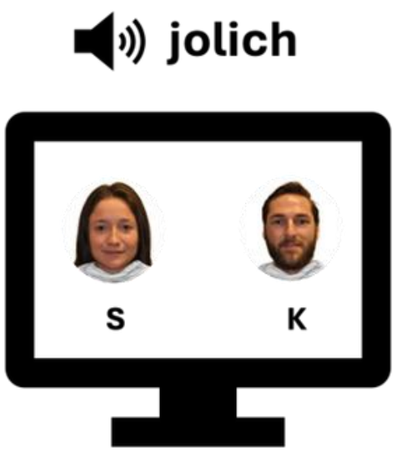
Examine which **phono-morphological cues** speakers use **for GG categorization** using nonwords. Experiment 1: explicit judgements. Experiment 2 & 3: implicit feature activation during reading.

3 Nonwords

20 Czech-like pseudo-stems + non-palatalized consonants (C, e.g., dy**r**); palatalized C (C', e.g., dy**ř**), C+ –a (e.g., dyr**a**), C+ –ka (e.g., dyr**ka**) and C+ –ek (e.g., dyr**ek**).

4 Methods

Experiment 1: Participants (N=128, 107 women, M_{age}=22.8, SD_{age}=4.4) saw a pair of faces (female and male), heard a novel word indicating a future profession and chose by keypress which face the word fit better.



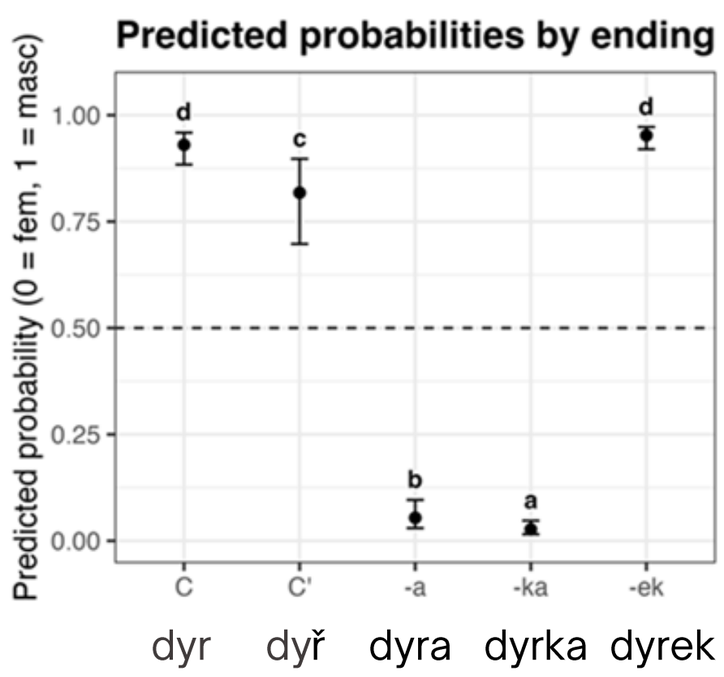
Experiment 2 (N=327; 263 women; M_{age}=22.6, SD_{age}=4.8) & Experiment 3 (N=364, 296 women; M_{age}=22.61, SD_{age}=4.4):

Participants read sentences in a self-paced reading task with moving-windows. Nonwords with -C, -a (Exp. 2) and –ek and –ka (Exp. 3) endings.

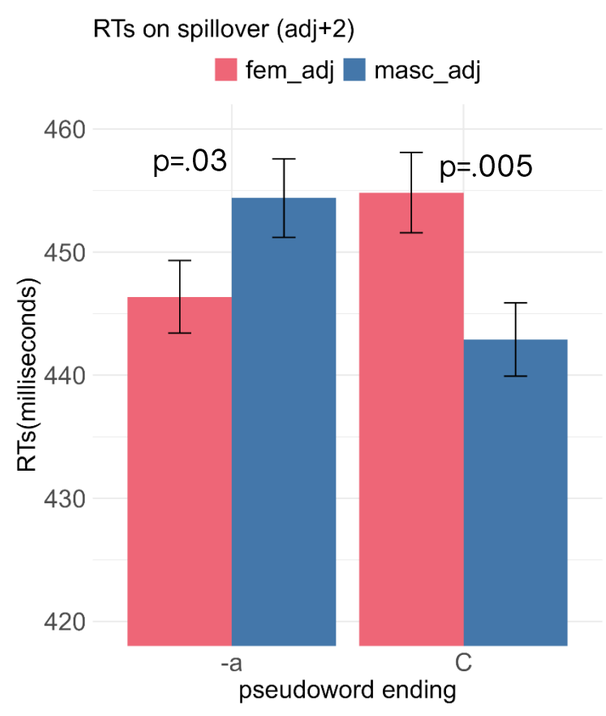
Region	Text	Gloss
intro	Zaznamenali jsme, že	We noticed that
nonword	**dyr**[**C**]/ **dyrek**[-**ek**]/ **dyra**[-**a**]/ **dyrka**[-**ka**]	nonword
copula	je	is
adj	**slepený**[M]/ **slepená**[F]	stuck together
spill	a	and
spill	nejde	cannot
wrap-up	rozlepit.	be separated

5 Results

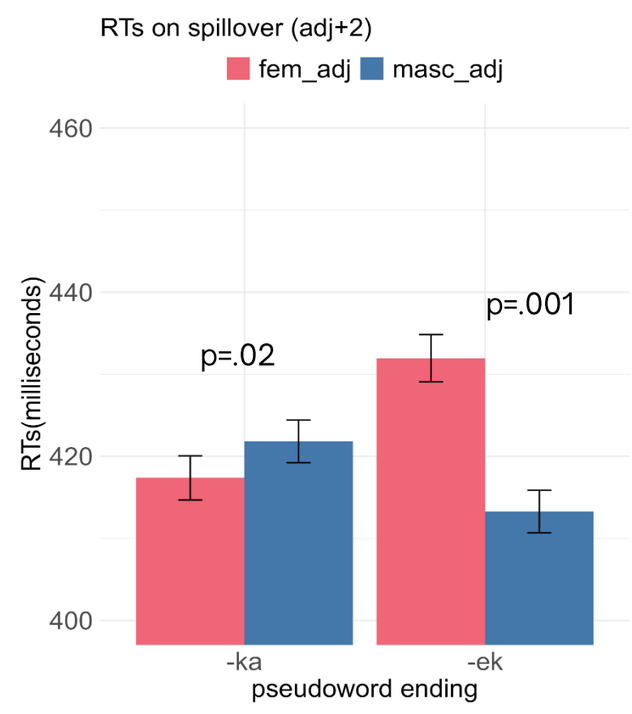
Experiment 1: glmer(response ~ ending +(1 + ending | participant) + (1 + ending | stem)



Experiment 2: lmer(RTs ~ ending + (1 + ending | participant) + (1 + ending | stem), data=spillover



Experiment 3: lmer(RTs ~ ending + (1 + ending | participant) + (1 | stem), data= spillover



6 Discussion

- Czech speakers **use word endings** to assign **grammatical gender** in both explicit (Exp 1) and implicit (Exp 2; Exp. 3) tasks.
- **-a** and **-ka** show a less clear pattern, likely due to greater morphological syncretism compared to **C**-final and **-ek** endings.
- Overall, speakers of Czech have a robust system of formal gender heuristics that operate independently of semantics.

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